Simple Stickers: An Interactive Multi-Model Image Segmentation Tool

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Abstract

In everyday life, people often need to isolate objects in photos—removing backgrounds from selfies, preparing product shots for online shops, or editing creative content. Current automatic tools often fail in cluttered backgrounds or with thin structures such as hair or wires. Manual segmentation, while accurate, is slow and requires skill.

We aim to create an ***interactive image segmentation tool*** *that makes object selection fast and accessible for everyday users by combining modern vision models with an intuitive interface.*
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Code: `https://github.com/adams-x0/cv_project/tree/fivemodels`

1. Introduction

Instance segmentation is an important task in computer vision where the goal is to detect objects in an image and assign a pixel-level mask to each individual object. Unlike simple object detection, which only produces bounding boxes, instance segmentation provides more detailed information about object shape and location. This makes it useful for real-world applications such as image editing, robotics, and interactive vision systems.

Although many modern segmentation models perform well on standard benchmarks, their performance can vary when applied to more challenging images. Real-world images often contain cluttered scenes, small or overlapping objects, unusual viewpoints, or low contrast between objects and the background. These factors can make it difficult for a single model to consistently produce accurate segmentation results.

In this project, we explore different segmentation approaches, and we evaluate multiple segmentation pipelines that combine object detection or grounding models with segmentation models. Faster approaches, such as YOLO-based segmentation, provide quick results but may struggle in complex scenes. More advanced models, such as OWL-ViT and SAM-based pipelines, can handle a wider variety of objects but introduce higher computational cost and latency. By comparing these methods, we aim to better understand their strengths, weaknesses, and suitability for interactive segmentation tasks.

2. Related Work

Instance segmentation has been an important research topic in computer vision because it combines object detection with pixel-level understanding. One of the most influential early approaches is Mask R-CNN [1], which introduced a two-stage pipeline that first detects objects using bounding boxes and then predicts a segmentation mask for each detected instance. While Mask R-CNN achieves strong segmentation accuracy, its multi-stage design makes it computationally expensive and less suitable for real-time or interactive applications.

To address speed limitations, more recent work has focused on single-stage, real-time segmentation models. The YOLO family of models extends real-time object detection by predicting segmentation masks directly in a single forward pass. Recent YOLO-based segmentation models, such as YOLOv12-Seg [2], prioritize low-latency inference and are commonly used in applications where responsiveness is critical. However, these speed-optimized models often sacrifice mask precision and semantic accuracy compared to heavier architectures.

Transformer-based approaches have also gained popularity in both detection and segmentation tasks. RT-DETR [4] introduces a real-time detection transformer that balances accuracy and speed through an efficient hybrid encoder and query-based detection without non-maximum suppression. While RT-DETR produces accurate bounding boxes, it does not directly output segmentation masks and must be combined with a separate segmentation model to obtain pixel-level results.

In contrast, Mask2Former [3] is a transformer-based universal segmentation model capable of performing semantic, instance, and panoptic segmentation within a single framework. Mask2Former produces high-quality segmentation masks but requires significantly more computation and offers less control over individual object selection.

More recently, foundation models and prompt-based segmentation methods have expanded segmentation beyond fixed label sets. Segment Anything (SAM) and its successor SAM2 [7] generate object masks based on prompts such as points or bounding boxes, enabling class-agnostic segmentation. Models such as GroundingDINO [6] and OWL-ViT [5] further extend this idea by incorporating natural-language prompts for open-vocabulary object detection. While these models provide strong flexibility, they typically rely on multi-stage pipelines that increase inference time and system complexity.

Most prior work evaluates segmentation models independently using benchmark datasets such as COCO, with limited emphasis on interactive usage or system-level comparison. In contrast, our project focuses on integrating multiple segmentation paradigms into a single interactive system, enabling direct qualitative comparison of speed, segmentation quality, and user controllability across different model classes.

3. Methods

3.1. System Overview

We developed *Simple Stickers*, an interactive image segmentation system that extracts objects from images and converts them into transparent "stickers" that can be reused and composited onto other images. The system integrates multiple state-of-the-art segmentation and detection models under a unified backend API, allowing direct comparison between real-time, transformer-based, prompt-based, and trained segmentation approaches.

The backend is implemented using FastAPI and exposes REST endpoints for each model, as well as a combined evaluation endpoint. The frontend is implemented in React (Vite) and provides an interactive interface for image upload, model selection, visualization of overlays, sticker extraction, and canvas-based composition.

3.2. Input Preprocessing and Image Handling

All input images are uploaded once and stored on the backend. Images are loaded using OpenCV or PIL depending on model requirements. To ensure consistent spatial alignment, EXIF orientation is corrected prior to inference for prompt-based pipelines, preventing portrait rotation artifacts. All images are processed at their original resolution to avoid resizing-induced segmentation errors.

3.3. YOLOv12 Segmentation

YOLOv12-Seg is used as a real-time instance segmentation baseline. The model directly predicts bounding boxes, class labels, confidence scores, and pixel-wise segmentation masks in a single forward pass.

For each detected object:

- The predicted binary mask is resized to match the original image resolution if necessary.
- A class-specific color overlay is blended with the original image for visualization.
- A transparent PNG "sticker" is generated by masking the background outside the object region.

This pipeline enables fast inference and is well suited for real-time applications.

3.4. Custom-Trained YOLOv12 Segmentation

In addition to the pretrained model, we evaluate a custom-trained YOLOv12-Seg model fine-tuned on a domain-specific dataset containing ten object categories. This allows evaluation of segmentation behavior on task-specific classes.

The inference pipeline is identical to the pretrained YOLOv12 approach, ensuring a fair comparison between generic and task-specific models.

3.5. RT-DETR with SAM2 Refinement

Real-Time Detection Transformer (RT-DETR) is used as a transformer-based object detector that outputs bounding boxes and class predictions but does not provide segmentation masks. To obtain high-quality object masks, each detected bounding box is passed to Segment Anything Model 2 (SAM2) for mask refinement.

The process consists of:

- RT-DETR bounding box detection.
- SAM2 mask prediction conditioned on each bounding box.
- Overlay visualization and transparent sticker extraction.

This two-stage approach enables evaluation of detection-only models augmented with high-quality segmentation.

3.6. Mask2Former Universal Segmentation

We integrate Mask2Former, a transformer-based universal segmentation model trained on COCO panoptic data. Mask2Former directly outputs panoptic segmentation maps containing both "thing" and "stuff" classes. Panoptic segmentation assigns a semantic label to every pixel in an image while simultaneously distinguishing individual object instances. Classes are divided into *things*, which represent countable object instances (e.g., person, car), and *stuff*, which represent uncountable background regions (e.g., sky, road).

Only instance ("thing") segments above a confidence threshold are retained. For each instance:

- The segmentation mask is extracted from the panoptic output.
- A bounding box is computed from the mask extent.
- A transparent sticker is generated using the extracted mask.

This pipeline enables class-agnostic instance extraction without explicit object detection.

3.7. Prompt-Based Segmentation Pipelines

3.7.1. GroundingDINO + SAM2

GroundingDINO is used to localize objects based on a natural-language prompt. The resulting bounding boxes are refined using SAM2 to obtain precise segmentation masks. This pipeline enables open-vocabulary, prompt-driven segmentation, making it suitable for interactive object search and extraction.

3.7.2. OWL-ViT + SAM2

OWL-ViT performs open-vocabulary object detection using vision–language embeddings. The highest-confidence detection matching the prompt is selected and refined with SAM2 to generate a segmentation mask. This approach provides a complementary prompt-based alternative and enables comparison between grounding-based and embedding-based detection strategies.

3.8. Backend API and Unified Evaluation

All models are exposed through a unified FastAPI backend. A dedicated `/segment/all` endpoint runs all segmentation pipelines on the same input image, enabling side-by-side qualitative comparison under identical conditions.

The outputs include:

- Overlay images with bounding boxes and segmentation masks.
- Individual transparent PNG stickers.
- Predicted class labels for each sticker.

This design ensures consistent evaluation and reproducibility across models.

3.9. Frontend Interface and Sticker Composition

The frontend allows users to:

- Select segmentation models.
- View overlays and extracted stickers.
- Download stickers.
- Composite stickers onto background images using a canvas-based editor with keyboard-controlled placement, scaling, and export.

This interface provides a complete end-to-end workflow from segmentation to creative reuse.

4. Experiments

4.1. Qualitative Comparison

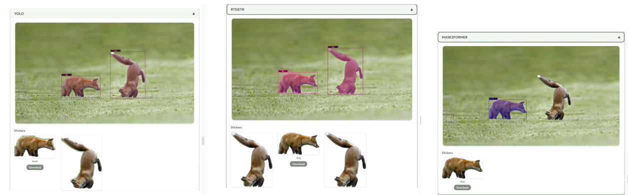
Figure 1 presents qualitative segmentation results on a challenging outdoor image containing two animals, one of which appears in an unusual inverted pose. This example evaluates instance separation, robustness to pose variation, and mask boundary quality.

YOLOv12 successfully detects and segments both animals, producing clean instance masks and well-separated stickers. However, it exhibits semantic misclassification by labeling both animals as "bear." Despite this, YOLO demonstrates strong instance recall and reliable separation, consistent with its detection-oriented design.

Mask2Former produces a high-quality segmentation mask for one animal but fails to detect the inverted instance. This behavior reflects its reliance on panoptic reasoning rather than explicit instance detection, leading to lower recall in complex poses while maintaining clean mask boundaries for detected objects.

RT-DETR combined with SAM2 detects both animals and produces high-fidelity segmentation masks. However, the refinement process occasionally results in over-segmentation, yielding duplicate stickers for the same object. While boundary quality is superior, additional post-processing is required to suppress redundant instances.

Overall, detection-based models favor reliable instance separation, universal segmentation models favor semantic consistency, and refinement-based pipelines favor mask quality at the cost of increased system complexity.



(a) YOLOv12 (b) RT-DETR + SAM2 (c) Mask2Former

Figure 1. Qualitative comparison on a challenging outdoor scene containing two foxes, including an inverted pose. YOLOv12 produces clean instance masks with reliable separation, RT-DETR + SAM2 achieves high-quality boundaries with occasional redundancy, and Mask2Former detects fewer instances in complex poses.

4.2. Qualitative Comparison on Crowded Scenes

Figure 2 shows qualitative segmentation results on a crowded outdoor scene containing multiple penguins with heavy occlusion and significant instance overlap.

RT-DETR achieves the highest instance recall in this example, detecting a larger number of penguins than the other models. Most visible individuals are localized, including partially occluded instances in the background. However, RT-DETR also produces redundant and overlapping detections, indicating that while recall is high, additional post-processing is required to suppress duplicate instances.

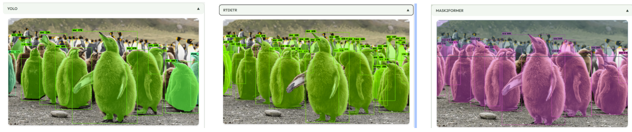
Mask2Former detects fewer penguins than RT-DETR in this crowded scene. Although the predicted masks are spatially consistent for detected instances, Mask2Former fails to segment several partially occluded penguins. This suggests that, in dense object layouts, Mask2Former's panoptic reasoning does not always translate into superior instance-level recall, particularly when object boundaries overlap significantly.

YOLOv12 detects the smallest number of penguins among the three models but produces relatively clean and well-separated instance masks for prominent individuals. Its conservative detection strategy leads to missed instances under occlusion but minimizes redundant predictions.

Overall, these results indicate that RT-DETR provides the strongest instance coverage in crowded scenes, followed by Mask2Former, while YOLOv12 prioritizes precision and instance separation over recall.

5. Conclusion

In this project, we presented *Simple Stickers*, an interactive image segmentation system that integrates five modern computer vision models into a unified pipeline for object extraction and reuse. By combining real-time instance seg-



(a) YOLOv12 (b) RT-DETR + SAM2 (c) Mask2Former

Figure 2. Qualitative comparison on a crowded scene of penguins with heavy occlusion and instance overlap. YOLOv12 produces clean but conservative instance masks, RT-DETR + SAM2 achieves the highest instance recall with some redundant detections, and Mask2Former detects fewer instances in dense regions.

mentation (YOLOv12), transformer-based detection (RT-DETR), universal panoptic segmentation (Mask2Former), and open-vocabulary prompt-based pipelines (GroundingDINO + SAM2 and OWL-ViT + SAM2), the system enables direct qualitative comparison across different segmentation paradigms within a single interface.

Our results demonstrate that segmentation performance strongly depends on both model architecture and interaction style. YOLOv12 provides fast and reliable segmentation for common object categories and is well suited for real-time use. The custom-trained YOLOv12 model further highlights the benefits of task-specific fine-tuning when operating within a known domain. Transformer-based detectors such as RT-DETR, when combined with SAM2, produce higher-quality object boundaries but introduce additional computational overhead. Mask2Former enables class-agnostic instance extraction by leveraging panoptic segmentation, though it offers less control for targeted object selection. Prompt-based pipelines provide the greatest flexibility by allowing users to segment arbitrary objects via natural language, but they are more sensitive to prompt phrasing and model confidence.

Beyond model comparison, this project emphasizes the importance of system-level design and usability in computer vision applications. The unified backend API, shared preprocessing steps, and the `/segment/all` endpoint ensure consistent evaluation across models, while the frontend enables interactive exploration, visualization, and creative reuse of segmentation outputs. By transforming segmentation results into reusable transparent "stickers," *Simple Stickers* demonstrates how computer vision systems can support both analytical and creative workflows.

Overall, this project illustrates that a multi-model, modular segmentation approach provides greater insight and flexibility than relying on a single method. It reflects current trends in computer vision toward combining pretrained models to support open-ended and interactive visual tasks.

6. Limitations

Despite its strengths, this project has several limitations.

First, computational cost and latency remain significant constraints. Models such as SAM2, Mask2Former, GroundingDINO, and OWL-ViT are computationally expensive, and pipelines that involve multiple stages (e.g., detection followed by segmentation) introduce noticeable delays. As a result, the full system is not suitable for real-time deployment without access to a powerful GPU.

Second, platform and hardware limitations restricted experimentation. Many state-of-the-art segmentation models require Linux-based CUDA environments to run efficiently or to access larger model variants. Development and testing on macOS limited the ability to evaluate these models under ideal conditions and constrained performance comparisons.

Third, evaluation was primarily qualitative and conducted on a limited set of images. While visual comparisons highlight differences in segmentation quality, robustness, and controllability, the absence of large-scale quantitative evaluation using labeled datasets (e.g., IoU, boundary accuracy, or panoptic metrics) limits the generalizability of the findings.

Finally, model selection and configuration are user-driven. The system does not currently include automated guidance or adaptive model selection based on image content or task intent. Users unfamiliar with the strengths and weaknesses of each model may select suboptimal pipelines, particularly when balancing speed, accuracy, and flexibility.

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